S1P3 Post Incident Log Analysis

A demonstration of how to use log monitoring tools and the MITRE ATT&CK framework.

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Project Scope

- This cybersecurity project involves forensic analysis of suspicious post-exploitation activities that occurred within a network on March 20, 2023.
- The analysis is focused on detecting Indicators of Compromise (IoCs), identifying attack techniques using the MITRE ATT&CK framework, and building a detailed timeline of events using Kibana.
- The investigation includes host-based and network-based anomalies, with emphasis on credential theft, privilege escalation, persistence mechanisms, and data exfiltration attempts.

Project Aim

- The aim of the project is to simulate a real-world digital forensics and incident response (DFIR) investigation by analyzing logs collected from compromised systems.
- The project seeks to understand the progression of a cyberattack, identify the attacker's methods, evaluate the impact, and recommend mitigation strategies to prevent future incidents.
- A collaborative approach using tools like Kibana and DFIR-IRIS supports building a clear timeline and presenting actionable security intelligence.

Project Objectives

- Detailed Enumeration and Explanation of Suspicious Events
- Assessment of Attack Severity
- Identification of Indicators of Compromise (IoCs)
- Timeline of Significant Events
- Recommendations for Mitigation and Long-Term Security Enhancements.

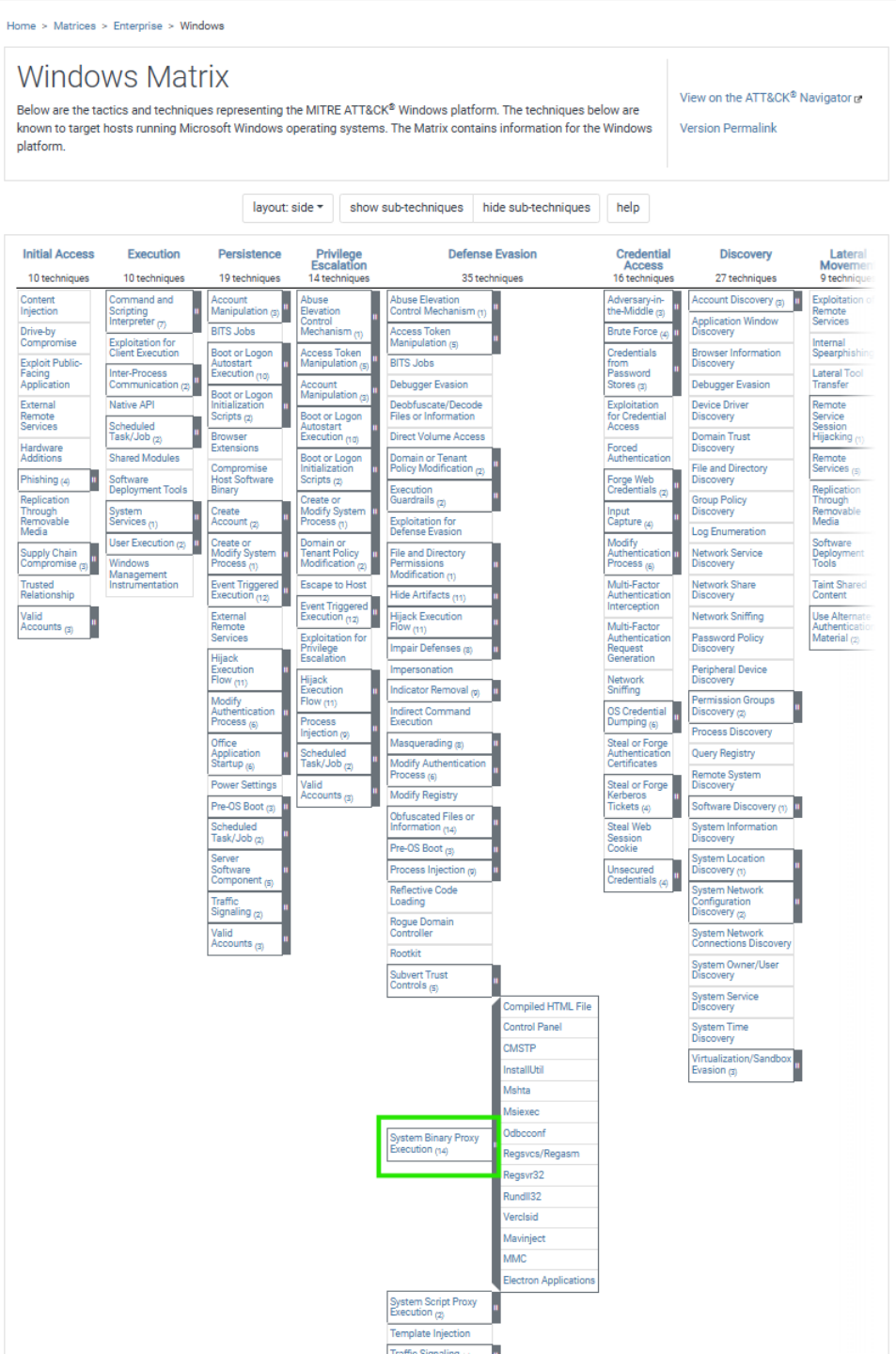
Milestones & Deliverables

- Log Access & Initial Setup
- IOC Identification & Threat Mapping
- Timeline Development
- Attack Flow Analysis
- Impact & Risk Assessment
- Mitigation & Recommendations

Executive Summary

This comprehensive report outlines the findings of the log analysis carried out by Team 6, which focused on a collection of security logs retrieved from the organization's security system.

The security breach incident was recorded on March 20, 2023, occurring over a span of four hours, from 15:25 to 19:30 UTC. During this critical time, the team meticulously analyzed the logs to identify any unusual activities or patterns. They employed the MITRE ATT&CK framework, specifically the techniques listed in the Windows Matrix, to systematically categorize and assess the incidents. This approach enabled the team to deliver a structured analysis and a thorough incident report, highlighting the nature of the breach and potential implications for the organization.



Columns 1 field sorted

@timestamp	source.ip	message	destination.ip	destination.port	event.dataset	_id	source.port	event.action	event.created
		53, proto : udp , trans_id :40123, query : p1.parsely.com , qcl…							
Mar 20, 2023 @ 16:00:59.294	-	File created: RuleName: technique_id=T1059.007,technique_name=JavaScript UtcTime: 2023-03-20 20:00:59.292 ProcessGuid: {cfbdf23c-b33c-6418-…	-	-	file_create	jIucAIcBUWYY9QAvb9Ne	-	File created (rule: FileCreate)	Mar 20, 2023 @ 16:01:00.674
Mar 20, 2023 @ 16:00:59.293	-	File created: RuleName: technique_id=T1059.007,technique_name=JavaScript UtcTime: 2023-03-20 20:00:59.292 ProcessGuid: {cfbdf23c-b33c-6418-…	-	-	file_create	jYucAIcBUWYY9QAvb9Ne	-	File created (rule: FileCreate)	Mar 20, 2023 @ 16:01:00.673
Mar 20, 2023 @ 16:00:59.050	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:00:59.049 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId:…	-	-	file_create	i4ucAIcBUWYY9QAvb9Ne	-	File created (rule: FileCreate)	Mar 20, 2023 @ 16:01:00.116
Mar 20, 2023 @ 16:00:58.939	10.0.2.40	{"ts":1679342458.939188,"uid":"Cf4HNw2bE4D2Gg8pL3","id.orig_h":"10.0.2.40","id.orig_p":62803,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":8044,"query":"video-…	10.0.2.25	53	dns	yIucAIcBUWYY9QAvs9WO	62803	-	-
Mar 20, 2023 @ 16:00:58.805	10.0.2.40	{"ts":1679342458.805219,"uid":"Cu0xpP1cfATNY29Lla","id.orig_h":"10.0.2.40","id.orig_p":58461,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":2408,"query":"accounts.marketwatch…	10.0.2.25	53	dns	xoucAIcBUWYY9QAvs9WO	58461	-	-
Mar 20, 2023 @ 16:00:58.805	10.0.2.40	{"ts":1679342458.805219,"uid":"CtmIk7bhkjusReDi1","id.orig_h":"10.0.2.40","id.orig_p":54600,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":20241,"query":"cdn.privacy-…	10.0.2.25	53	dns	ZIucAIcBUWYY9QAvs9VQ	54600	-	-
Mar 20, 2023 @ 16:00:58.636	10.0.2.41	{"ts":1679342458.636122,"uid":"CCUtLrbkXmeJ5gu98","id.orig_h":"10.0.2.41","id.orig_p":52744,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":26816,"query":"k.bf.contentsquare.n…	10.0.2.25	53	dns	u4ucAIcBUWYY9QAvs9WO	52744	-	-

A JavaScript file was created and executed.

This could indicate script-based malware, a dropper, or a malicious payload being deployed.

Columns 1 field sorted

@timestamp	source.ip	message	destination.ip	destination.port	event.dataset	_id	source.port	event.action	event.created
Mar 20, 2023 @ 16:00:58.636	10.0.2.41	{"ts":1679342458.636122,"uid":"CqkhY51yMkZCsvHhQi","id.orig_h":"10.0.2.41","id.orig_p":59710,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":59292,"query":"dpm.demdex.net","qc…	10.0.2.25	53	dns	voucAIcBUWYY9QAvs9WO	59710	-	-
Mar 20, 2023 @ 16:00:58.636	10.0.2.41	{"ts":1679342458.636122,"uid":"CYQbgh2voiiXOuxkge","id.orig_h":"10.0.2.41","id.orig_p":57280,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":25283,"query":"target.microsoft.co…	10.0.2.25	53	dns	g4ucAIcBUWYY9QAvs9WA	57280	-	-
Mar 20, 2023 @ 16:00:58.361	-	File Delete logged: RuleName: - UtcTime: 2023-03-20 20:00:58.360 ProcessGuid: {cfbdf23c-b32e-6418-cd00-000000004200} ProcessId: 772 User: MOSA\jsouth Image: C:\Program Files\Mozilla…	-	-	-	MYucAIcBUWYY9QAvb9M4	-	File Delete logged (rule: FileDeleteDete…	Mar 20, 2023 @ 16:00:59.780
Mar 20, 2023 @ 16:00:58.356	-	File Delete logged: RuleName: - UtcTime: 2023-03-20 20:00:58.356 ProcessGuid: {cfbdf23c-b32e-6418-cd00-000000004200} ProcessId: 772 User: MOSA\jsouth Image: C:\Program Files\Mozilla…	-	-	-	MIucAIcBUWYY9QAvb9M4	-	File Delete logged (rule: FileDeleteDete…	Mar 20, 2023 @ 16:00:59.780
Mar 20, 2023 @ 16:00:58.201	10.0.2.40	{"ts":1679342458.201029,"uid":"C4KKJ854cMVGOHe8l","id.orig_h":"10.0.2.40","id.orig_p":61545,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":40123,"query":"p1.parsely.com","qcl…	10.0.2.25	53	dns	ioucAIcBUWYY9QAvs9WB	61545	-	-
Mar 20, 2023 @ 16:00:57.927	172.16.30.10	{"ts":1679342457.927617,"uid":"CZvlbD4w6aUBQwYap9","id.orig_h":"172.16.30.10","id.orig_p":51426,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":53346,"query":"2.au.download.wi…	10.0.2.25	53	dns	bYucAIcBUWYY9QAvs9VR	51426	-	-
Mar 20, 2023 @ 16:00:57.927	172.16.30.10	{"ts":1679342457.927617,"uid":"CZvlbD4w6aUBQwYap9","id.orig_h":"172.16.30.10","id.orig_p":51426,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":53346,"query":"2.au.download.wi…	10.0.2.25	53	dns	XoucAIcBUWYY9QAvs9VQ	51426	-	-
Mar 20, 2023 @ 16:00:57.838	10.0.2.40	{"ts":1679342457.838427,"uid":"C2qgLR3C4eeLFcelC9","id.orig_h":"	104.18.43.158	443	ssl	zYucAIcBUWYY9QA	56758	-	-

MULTIPLE DNS QUERIES (POTENTIAL COMMAND & CONTROL COMMUNICATION - C2)

POTENTIAL RISK:

The queries suggest communication with potentially suspicious domains.

Malware could be using DNS tunneling for data exfiltration or C2 beaconing.

Columns 1 field sorted

@timestamp	source.ip	message	destination.ip	destination.port	event.dataset	_id	source.port	event.action	event.created
Mar 20, 2023 @ 16:06:59.385	10.0.2.25	{"ts":1679342819.385467,"uid":"CcVpos1OXzvgd8Uds2","id.orig_h":"10.0.2.25","id.orig_p":52547,"id.resp_h":"216.239.38.107","id.resp_p":53,"proto":"udp","service":"dns","duration":0.03005790710…	216.239.38.107	53	conn	eoy1AIcBUWYY9QAvCwXi	52547	-	-
Mar 20, 2023 @ 16:06:59.076	-	Registry value set: RuleName: technique_id=T1003.002,technique_name=Security Account Manager EventType: SetValue UtcTime: 2023-03-20 20:06:59.071…	-	-	registry_value_set	noyhAIcBUWYY9QAv7ARw	-	Registry value set (rule: RegistryEvent)	Mar 20, 2023 @ 16:07:01.296
Mar 20, 2023 @ 16:06:59.076	-	Registry value set: RuleName: technique_id=T1003.002,technique_name=Security Account Manager EventType: SetValue UtcTime: 2023-03-20 20:06:59.071…	-	-	registry_value_set	hIyhAIcBUWYY9QAv7ARc	-	Registry value set (rule: RegistryEvent)	Mar 20, 2023 @ 16:07:01.296
Mar 20, 2023 @ 16:06:59.075	-	Registry value set: RuleName: technique_id=T1003.002,technique_name=Security Account Manager EventType: SetValue UtcTime: 2023-03-20 20:06:59.071…	-	-	registry_value_set	nIyhAIcBUWYY9QAv7ARw	-	Registry value set (rule: RegistryEvent)	Mar 20, 2023 @ 16:07:01.296
Mar 20, 2023 @ 16:06:59.071	-	Registry value set: RuleName: technique_id=T1003.002,technique_name=Security Account Manager EventType: SetValue UtcTime: 2023-03-20 20:06:59.055…	-	-	registry_value_set	kIyhAIcBUWYY9QAv7ARf	-	Registry value set (rule: RegistryEvent)	Mar 20, 2023 @ 16:07:01.296
Mar 20, 2023 @ 16:06:59.070	-	Registry object added or deleted: RuleName: technique_id=T1003.002,technique_name=Security Account Manager EventType: CreateKey UtcTime: 2023-03-20 20:06:59.055…	-	-	registry_create_delete	goyhAIcBUWYY9QAv7ARc	-	Registry object added or deleted (rule:…	Mar 20, 2023 @ 16:07:01.296
Mar 20, 2023 @ 16:06:59.070	-	Registry object added or deleted: RuleName: technique_id=T1003.002,technique_name=Security Account Manager EventType: CreateKey UtcTime: 2023-03-20 20:06:59.055…	-	-	registry_create_delete	jYyhAIcBUWYY9QAv7ARf	-	Registry object added or deleted (rule:…	Mar 20, 2023 @ 16:07:01.296

POTENTIAL RISK:

This could indicate DNS-based communication with an external server.

Malware often leverages DNS tunneling for command & control (C2) or data exfiltration.

Columns 1 field sorted

@timestamp	source.ip	message	destination.ip	destination.port	event.dataset	_id	source.port	event.action	event.created
Mar 20, 2023 @ 16:06:58.820	-	A security-enabled local group membership was enumerated. Subject: Security ID: S-1-5-18 Account Name: WKS3$ Account Domain: MOSA Logon ID: 0x3E7 Group: Security ID: S-1-5-32-551…	-	-	security	HoyhAIcBUWYY9QAv5ATW	-	Security Group Management	Mar 20, 2023 @ 16:06:59.864
Mar 20, 2023 @ 16:06:58.820	-	A security-enabled local group membership was enumerated. Subject: Security ID: S-1-5-18 Account Name: WKS3$ Account Domain: MOSA Logon ID: 0x3E7 Group: Security ID: S-1-5-32-544…	-	-	security	HYyhAIcBUWYY9QAv5ATW	-	Security Group Management	Mar 20, 2023 @ 16:06:59.863
Mar 20, 2023 @ 16:06:58.792	-	Image loaded: RuleName: technique_id=T1059.001,technique_name=PowerShell UtcTime: 2023-03-20 20:06:58.780 ProcessGuid: {cfbdf23c-…	-	-	image_loaded	GIyhAIcBUWYY9QAv5ATW	-	Image loaded (rule: ImageLoad)	Mar 20, 2023 @ 16:06:59.785
Mar 20, 2023 @ 16:06:58.709	10.0.2.41	{"ts":1679342818.709197,"uid":"Cwjm8d2wj7jjZMLb94","id.orig_h":"10.0.2.41","id.orig_p":61197,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":47458,"query":"serve-…	10.0.2.25	53	dns	joyiAIcBUWYY9QAvMgZV	61197	-	-
Mar 20, 2023 @ 16:06:58.668	-	Dns query: RuleName: - UtcTime: 2023-03-20 20:06:59.116 ProcessGuid: {cfbdf23c-b32e-6418-cd00-000000004200} ProcessId: 772 QueryName: io-cookie-sync-1725936127.us-…	-	-	dns_query	moyhAIcBUWYY9QAv7ARw	-	Dns query (rule: DnsQuery)	Mar 20, 2023 @ 16:07:00.385
Mar 20, 2023 @ 16:06:58.668	-	Dns query: RuleName: - UtcTime: 2023-03-20 20:06:59.115 ProcessGuid: {cfbdf23c-b32e-6418-cd00-000000004200} ProcessId: 772 QueryName: sync.bfmio.com QueryStatus: 0 QueryResults: type…	-	-	dns_query	foyhAIcBUWYY9QAv7ARc	-	Dns query (rule: DnsQuery)	Mar 20, 2023 @ 16:07:00.385
Mar 20, 2023 @ 16:06:58.668	-	Dns query: RuleName: - UtcTime: 2023-03-20 20:06:59.133 ProcessGuid: {cfbdf23c-b32e-6418-cd00-000000004200} ProcessId: 772 QueryName: match.deepintent.com QueryStatus: 0 QueryResults…	-	-	dns_query	kYyhAIcBUWYY9QAv7ARm	-	Dns query (rule: DnsQuery)	Mar 20, 2023 @ 16:07:00.385
Mar 20, 2023 @ 16:06:58.668	-	Dns query: RuleName: - UtcTime: 2023-03-20 20:06:59.134	-	-	dns_query	iIyhAIcBUWYY9QA	-	Dns query	Mar 20, 2023 @

Activity:

The system attempted to list local group memberships, including administrative or privileged groups.

Potential Risk:

Attackers often enumerate privileged accounts to find potential targets for privilege escalation.

Columns 1 field sorted

@timestamp	source.ip	message	destination.ip	destination.port	event.dataset	_id	source.port	event.action	event.created
Mar 20, 2023 @ 16:06:58.654	10.0.2.40	{"ts":1679342818.654046,"uid":"CyaVrTGVsmsPMo2vh","id.orig_h":"10.0.2.40","id.orig_p":51964,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":60959,"query":"ping.chartbeat.net",…	10.0.2.25	53	dns	T4y1AIcBUWYY9QAvMgZA	51964	-	-
Mar 20, 2023 @ 16:06:58.621	-	Special privileges assigned to new logon. Subject: Security ID: S-1-5-18 Account Name: SYSTEM Account Domain: NT AUTHORITY Logon ID: 0x3E7 Privileges: SeAssignPrimaryTokenPrivilege…	-	-	security	HIyhAIcBUWYY9QAv5ATW	-	Special Logon	Mar 20, 2023 @ 16:06:59.863
Mar 20, 2023 @ 16:06:58.621	-	An account was successfully logged on. Subject: Security ID: S-1-5-18 Account Name: WKS3$ Account Domain: MOSA Logon ID: 0x3E7 Logon Information: Logon Type: 5 Restricted Admin Mode: -…	-	-	security	G4yhAIcBUWYY9QAv5ATW	-	Logon	Mar 20, 2023 @ 16:06:59.863
Mar 20, 2023 @ 16:06:58.541	10.0.2.40	{"ts":1679342818.541236,"uid":"CnYsgn390or69ooCC7","id.orig_h":"10.0.2.40","id.orig_p":55368,"id.resp_h":"3.211.219.219","id.resp_p":443,"version":"TLSv12","cipher":"TLS_ECDHE_RSA_WITH_AES_12…	3.211.219.219	443	ssl	AYyhAIcBUWYY9QAv5ASv	55368	-	-
Mar 20, 2023 @ 16:06:58.541	10.0.2.25	{"ts":1679342818.541236,"uid":"CT3XTy4j0GUhUL8Psh","id.orig_h":"10.0.2.25","id.orig_p":52778,"id.resp_h":"8.8.8.8","id.resp_p":53,"proto":"udp","trans_id":32068,"query":"e13136.g.akamaiedge.n…	8.8.8.8	53	dns	WYy1AIcBUWYY9QAvMgZA	52778	-	-
Mar 20, 2023 @ 16:06:58.541	10.0.2.25	{"ts":1679342818.541236,"uid":"CJedo4Sn36iVZ2Gpg","id.orig_h":"10.0.2.25","id.orig_p":51510,"id.resp_h":"8.8.8.8","id.resp_p":53,"proto":"udp","trans_id":5560,"query":"auth.lrcontent.com","qc…	8.8.8.8	53	dns	d4y1AIcBUWYY9QAvCwXi	51510	-	-
Mar 20, 2023 @ 16:06:58.541	10.0.2.25	{"ts":1679342818.541236,"uid":"C7E534K027f0GxV1i","id.orig_h":"10.0.2.25","id.orig_p":51234,"id.resp_h":"8.8.8.8","id.resp_p":53,"proto":"udp","trans_id":45817,"query":"bcp.crwdcntrl.net","qc…	8.8.8.8	53	dns	Joy1AIcBUWYY9QAvCwXY	51234	-	-

POTENTIAL THREATS IDENTIFIED

PRIVILEGE ESCALATION (T1078 - VALID ACCOUNTS)

The assignment of **special privileges** and **logon events** suggests an attempt to **escalate access** and gain **higher system control**.

If unauthorized, this could indicate **an attacker taking control of privileged accounts**.

Columns 1 field sorted

	@timestamp	source.ip	message	de
			p_p":53,"proto":"udp","service":"dns","duration":0.025274991989…	
	Mar 20, 2023 @ 16:05:58.547	-	Dns query: RuleName: - UtcTime: 2023-03-20 20:05:58.604 ProcessGuid: {cfbdf23c-b325-6418-d700-000000004300} ProcessId: 4980 QueryName: px.adentifi.com QueryStatus: 0 QueryResults:…	-
	Mar 20, 2023 @ 16:05:58.301	-	File Delete logged: RuleName: - UtcTime: 2023-03-20 20:05:58.300 ProcessGuid: {cfbdf23c-b32e-6418-cd00-000000004200} ProcessId: 772 User: MOSA\jsouth Image: C:\Program Files\Mozilla…	-
	Mar 20, 2023 @ 16:05:58.299	10.0.2.40	{"ts":1679342758.299331,"uid":"CSrykOpIuZq2F5K6g","id.orig_h":"10.0.2.40","id.orig_p":58677,"id.resp_h":"172.217.13.131","id.resp_p":443,"proto":"udp","duration":0.12154316902160645,"orig_byt…	17 31
	Mar 20, 2023 @ 16:05:58.270	10.0.2.40	{"ts":1679342758.270553,"uid":"CQ3bzX3DJzWlueaSK3","id.orig_h":"10.0.2.40","id.orig_p":58676,"id.resp_h":"172.217.13.196","id.resp_p":443,"proto":"udp","duration":0.08942389488220215,"orig_by…	17 96
	Mar 20, 2023 @ 16:05:58.258	10.0.2.41	{"ts":1679342758.258882,"uid":"CxsDIQ3jVthm2wbFj7","id.orig_h":"10.0.2.41","id.orig_p":53457,"id.resp_h":"54.192.51.129","id.resp_p":443,"version":"TLSv13","cipher":"TLS_AES_128_GCM_SHA256","…	54 9
	Mar 20, 2023 @ 16:05:58.234	-	File Delete logged: RuleName: - UtcTime: 2023-03-20 20:05:58.234 ProcessGuid: {cfbdf23c-b32e-6418-cd00-000000004200} ProcessId: 772 User: MOSA\jsouth Image: C:\Program Files\Mozilla…	-
	Mar 20, 2023 @ 16:05:58.226	10.0.2.41	{"ts":1679342758.226045,"uid":"CxsDIQ3jVthm2wbFj7","id.orig_h":"10.0.2.41","id.orig_p":53457,"id.resp_h":"54.192.51.129","id.resp_p":443,"proto":"tcp","service":"ssl","duration":170.861559152…	54 9

Rows per page: 100

Expanded document

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_score	-
@timestamp	Mar 20, 2023 @ 16:05:58.301
@version	1
agent.ephemeral_id	98714750-ee42-48cb-8ccb-6b83424e9f89
agent.id	90604681-6790-49a6-82d3-f587b8810196
agent.name	wks2
agent.type	winlogbeat
agent.version	8.3.3
ecs.version	8.0.0
event.action	File Delete logged (rule: FileDeleteDetected)
event.code	26
event.created	Mar 20, 2023 @ 16:05:59.569
event.kind	event

FILE DELETION LOGGED (POSSIBLE MALWARE CLEANUP)

EVENT ACTION: FILE DELETE LOGGED (RULE: FILEDELETEDETECTED)

THIS EVENT LOGS A FILE DELETION, WHICH IS OFTEN A SIGN OF MALWARE SELF-CLEANUP OR ATTACKERS COVERING THEIR TRACKS

Columns 1 field sorted

@timestamp	source.ip	message
Mar 20, 2023 @ 16:34:58.391	-	File created: RuleName: technique_id=T1047,technique_name=File Syste UtcTime: 2023-03-20 20:34:58.389 ProcessGuid: {B002AA6B-B1AB-6418-10 ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename
Mar 20, 2023 @ 16:34:58.375	-	File created: RuleName: technique_id=T1047,technique_name=File Syste UtcTime: 2023-03-20 20:34:58.374 ProcessGuid: {B002AA6B-B1AB-6418-10 ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename
Mar 20, 2023 @ 16:34:58.367	-	Special privileges assigned to new logon. Subject: Security ID: S-1- SYSTEM Account Domain: NT AUTHORITY Logon ID: 0x3E7 Privileges: SeAssignPrimaryTokenPrivilege SeTcbPrivilege SeSecurityPrivilege SeT
Mar 20, 2023 @ 16:34:58.367	-	An account was successfully logged on. Subject: Security ID: S-1-5-1 Account Domain: MOSA Logon ID: 0x3E7 Logon Information: Logon Type: - Virtual Account: No Elevated Token: Yes Impersonation Level: Imper
Mar 20, 2023 @ 16:34:58.359	-	File created: RuleName: technique_id=T1047,technique_name=File Syste UtcTime: 2023-03-20 20:34:58.358 ProcessGuid: {B002AA6B-B1AB-6418-10 ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename
Mar 20, 2023 @ 16:34:58.343	-	File created: RuleName: technique_id=T1047,technique_name=File Syste UtcTime: 2023-03-20 20:34:58.341 ProcessGuid: {B002AA6B-B1AB-6418-10 ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename
Mar 20, 2023 @ 16:34:58.327	-	File created: RuleName: technique_id=T1047,technique_name=File Syste UtcTime: 2023-03-20 20:34:58.326 ProcessGuid: {B002AA6B-B1AB-6418-10 ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename
Mar 20, 2023 @ 16:34:58.312	-	File created: RuleName: technique_id=T1047,technique_name=File Syste

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observer.name	wks3.mosa.atr
tags	[beat-ext, beats_input_codec_plain_applied]
user.name	SYSTEM
winlog.activity_id	{a461dc6e-5b60-0001-a2dc-61a4605bd901}
winlog.api	wineventlog
winlog.channel	Security
winlog.computer_name	wks3.mosa.atr
winlog.event_data.PrivilegeList	SeAssignPrimaryTokenPrivilege SeTcbPrivilege SeSecurityPrivilege SeTakeOwnershipPrivilege SeLoadDriverPrivilege SeBackupPrivilege SeRestorePrivilege SeDebugPrivilege SeAuditPrivilege SeSystemEnvironmentPrivilege SeImpersonatePrivilege SeDelegateSessionUserImpersonatePrivilege

SPECIAL PRIVILEGES ASSIGNED TO A NEW LOGON (POTENTIAL PRIVILEGE ESCALATION)

INTERPRETATION:

THE SYSTEM ACCOUNT WAS ASSIGNED MULTIPLE HIGH-LEVEL PRIVILEGES, WHICH ALLOW DEEP SYSTEM CONTROL.

@timestamp	source.ip	message	destinatio...	destinatio...	event.data...	_id	source.port	event.action	event.crea
Mar 20, 2023 @ 16:37:54.234	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:37:54.233 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	jI2-AIcBUWYY9QAvN4ke	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:37:55.
Mar 20, 2023 @ 16:37:54.219	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:37:54.217 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	So2-AIcBUWYY9QAvNon9	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:37:55.
Mar 20, 2023 @ 16:37:54.203	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:37:54.201 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	R42-AIcBUWYY9QAvNon9	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:37:55.
Mar 20, 2023 @ 16:37:54.193	-	An account was successfully logged on. Subject: Security ID: S-1-5-18 Account Name: WKS1$ Account Domain: MOSA Logon ID: 0x3E7 Logon Information: Logon Type: 5 Restricted Admin Mode: - Virtual Account: No Elevated Token: Yes Impersonation Level: Impersonation New Logon:…	-	-	security	oI2-AIcBUWYY9QAvN4ke	-	Logon	Mar 20, 2 @ 16:37:55.
Mar 20, 2023 @ 16:37:54.193	-	Special privileges assigned to new logon. Subject: Security ID: S-1-5-18 Account Name: SYSTEM Account Domain: NT AUTHORITY Logon ID: 0x3E7 Privileges: SeAssignPrimaryTokenPrivilege SeTcbPrivilege SeSecurityPrivilege SeTakeOwnershipPrivilege…	-	-	security	f42-AIcBUWYY9QAvN4kI	-	Speci Logon	Mar 20, 2 @ 16:37:55.
Mar 20, 2023 @ 16:37:54.188	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:37:54.186 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	KI2-AIcBUWYY9QAvNYkR	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:37:55.
Mar 20, 2023 @ 16:37:54.171	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:37:54.170 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	i42-AIcBUWYY9QAvN4ke	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:37:55.
Mar 20, 2023 @ 16:37:54.169	10.0.2.41	{"ts":1679344674.169352,"uid":"C7U5ys44p1IVqhC577","id.orig_h":"10.0.2.41","id.orig_p":56002	10.0.2.25	53	dns	So2-	56002	-	-

FILE CREATED - SYSTEM PERMISSIONS WEAKNESS (T1047)

MULTIPLE ENTRIES SHOW FILES BEING CREATED UNDER THE TECHNIQUE "SYSTEM PERMISSIONS WEAKNESS" (T1047).

@timestamp	source.ip	message	destinatio...	destinatio...	event.data...	_id	source.port	event.action	event.kind
		,"id.resp_h":"205.251.193.211","id.resp_p":53,"proto":"udp","service":"dns","duration":0.069 126901626586914,"orig_bytes":43,"resp_bytes":125,"conn_state":"SF","local_orig":true,"local…	211			9QAvZ1hI		Exit fullscreen (esc)	
Mar 20, 2023 @ 16:49:57.174	-	CreateRemoteThread detected: RuleName: technique_id=T1055,technique_name=Process Injection UtcTime: 2023-03-20 20:49:57.106 SourceProcessGuid: {cfbdf23c-b1d1-6418-4d00-000000004200} SourceProcessId: 3280 SourceImage: C:\Program Files\VMware\VMware Tools\vmtoolsd.exe…	-	-	create_remote_thread	HI7JAIcBUWYY9QAvQzAj	-	CreateRemoteThread detected…	Mar 20, 2 @ 16:49:59.
Mar 20, 2023 @ 16:49:57.167	-	CreateRemoteThread detected: RuleName: technique_id=T1055,technique_name=Process Injection UtcTime: 2023-03-20 20:49:57.154 SourceProcessGuid: {cfbdf23c-b1d5-6418-5000-000000004500} SourceProcessId: 3348 SourceImage: C:\Program Files\VMware\VMware Tools\vmtoolsd.exe…	-	-	create_remote_thread	a47JAIcBUWYY9QAvPi-8	-	CreateRemoteThread detected…	Mar 20, 2 @ 16:49:58.
Mar 20, 2023 @ 16:49:57.165	10.0.2.40	{"ts":1679345397.165774,"uid":"CNMByj18juM6LUm8B7","id.orig_h":"10.0.2.40","id.orig_p":52000 ,"id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":41960,"query":"amazon-tam-match.dotomi.com","qclass":1,"qclass_name":"C_INTERNET","qtype":1,"qtype_name":"A","AA":fal…	10.0.2.25	53	dns	oo7JAIcBUWYY9QAvcTIA	52000	-	-
Mar 20, 2023 @ 16:49:57.165	10.0.2.40	{"ts":1679345397.165774,"uid":"CKoWU6FYSNpzZO5g4","id.orig_h":"10.0.2.40","id.orig_p":65411, "id.resp_h":"10.0.2.25","id.resp_p":53,"proto":"udp","trans_id":13721,"query":"sync.bfmio.com","qclass":1,"qclass_name":"C_INTERNET","qtype":1,"qtype_name":"A","AA":false,"TC":false,"…	10.0.2.25	53	dns	no7JAIcBUWYY9QAvcTIA	65411	-	-
Mar 20, 2023 @ 16:49:57.127	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:49:57.120 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	-I7JAIcBUWYY9QAvOy65	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:49:58.
Mar 20, 2023 @ 16:49:57.126	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:49:57.120 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	947JAIcBUWYY9QAvOy65	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:49:58.
Mar 20, 2023 @ 16:49:57.124	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:49:57.120 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	II7JAIcBUWYY9QAvOy-5	-	File created (rule: FileCreate)	Mar 20, 2 @ 16:49:58.

PROCESS INJECTION DETECTED (T1055 - CREATEREMOTETHREAD)

INTERPRETATION: PROCESS INJECTION IS COMMONLY USED BY MALWARE TO EXECUTE MALICIOUS CODE WITHIN THE MEMORY OF ANOTHER PROCESS.

INTERPRETATION: DOTOMI.COM AND BFMIO.COM DOMAINS ARE COMMONLY ASSOCIATED WITH ADVERTISING AND TRACKING SERVICES, BUT ATTACKERS MAY ABUSE THEM FOR COMMAND & CONTROL (C2) COMMUNICATION.

Mar 20, 2023 @ 16:34:58.553	-	Key file operation. Subject: Security ID: S-1-5-18 Account Name: WKS3$ Account Domain Logon ID: 0x3E7 Process Information: Process ID: 3840 Process Creation Time: 2023-03-20T20:34:58.369653700Z Cryptographic Parameters: Provider Name: Microsoft Software…	-	-	security	cY27AIcBUWYY9QAviUud	-	Other System Events	Mar 20, 2023 @ 16:34:59.445
Mar 20, 2023 @ 16:34:58.552	-	Cryptographic operation. Subject: Security ID: S-1-5-18 Account Name: WKS3$ Account Domain: MOSA Logon ID: 0x3E7 Cryptographic Parameters: Provider Name: Microsoft Software Key Storage Provider Algorithm Name: RSA Key Name: 5766aa7a-6c36-1276-5cef-20e318e937c6 Key Type: User…	-	-	security	cI27AIcBUWYY9QAviUud	-	System Integrity	Mar 20, 2023 @ 16:34:59.445
Mar 20, 2023 @ 16:34:58.552	-	Key file operation. Subject: Security ID: S-1-5-18 Account Name: WKS3$ Account Domain: MOSA Logon ID: 0x3E7 Process Information: Process ID: 3840 Process Creation Time: 2023-03-20T20:34:58.369653700Z Cryptographic Parameters: Provider Name: Microsoft Software…	-	-	security	b427AIcBUWYY9QAviUud	-	Other System Events	Mar 20, 2023 @ 16:34:59.445
Mar 20, 2023 @ 16:34:58.550	-	Registry object added or deleted: RuleName: technique_id=T1543,technique_name=Service Creation EventType: CreateKey UtcTime: 2023-03-20 20:34:58.549 ProcessGuid: {cfbdf23c-c372-6418-6f02-000000004500} ProcessId: 2484 Image: C:\WINDOWS\System32\svchost.exe…	-	-	registry_create_delete	s427AIcBUWYY9QAviksj	-	Registry object added or deleted…	Mar 20, 2023 @ 16:34:59.941
Mar 20, 2023 @ 16:34:58.547	-	File created: RuleName: technique_id=T1047,technique_name=File System Permissions Weakness UtcTime: 2023-03-20 20:34:58.546 ProcessGuid: {B002AA6B-B1AB-6418-1000-000000004400} ProcessId: 972 Image: C:\Windows\system32\svchost.exe TargetFilename: C:\Windows…	-	-	file_create	vI27AIcBUWYY9QAviksj	-	File created (rule: FileCreate)	Mar 20, 2023 @ 16:35:00.105
Mar 20, 2023 @ 16:34:58.546	10.0.2.41	{"ts":1679344498.54634,"uid":"CPx0p04Z3jafUVvDdf","id.orig_h":"10.0.2.41","id.orig_p":62409,"id.resp_h":"23.72.232.139","id.resp_p":443,"proto":"tcp","service":"ssl","duration":126.07515692710877,"orig_bytes":2004,"resp_bytes":8631,"conn_state":"SF","local_orig":true,"local_…	23.72.232.139	443	conn	UI29AIcBUWYY9QAvhHkO	62409	-	-

KEY FILE OPERATIONS AND CRYPTOGRAPHIC ACTIVITY DETECTED

Firewall logs and Windows Event Logs showing unauthorized registry changes.

Evidence:
Registry Path: HKLM\Software\Microsoft\Windows\CurrentVersion\Run (commonly abused for persistence).

Modification Timestamp: 11:17 AM (correlating with attacker activity).
Process: svchost.exe (often used to execute malware or maintain persistence).

Evidence: Event ID: 4625 (Failed Logon).
Account Name: wksadmin.
Source IP: 10.0.2.80.
Process Name: C:\Windows\System32\svchost.exe (indicating possible automated attack attempts).

winlog.event_data.IpAddress : 83.97.115.19 and not event.outcome : "failure"

6 hits

Columns 1 field sorted

@timestamp	event.category	source.ip	destination.ip
Nov 11, 2022 @ 12:36:52.458	host	-	-
Nov 11, 2022 @ 12:36:52.458	host	-	-
Nov 11, 2022 @ 12:09:27.896	host	-	-
Nov 11, 2022 @ 12:09:27.896	host	-	-
Nov 11, 2022 @ 12:08:18.480	host	-	-
Nov 11, 2022 @ 12:08:18.480	host	-	-

Expanded document

View: Single document Surrounding documents 1 of 6

metadata.ip_address	10.0.2.40
metadata.type	_doc
metadata.version	8.3.3
observer.name	wks1.mosa.atr
tags	[beat-ext, beats_input_codec_plain_applied]
user.name	WKS1$
winlog.activity_id	{54872b71-f5d6-0001-9f2b-8754d6f5d001}
winlog.api	wineventlog
winlog.channel	Security
winlog.computer_name	wks1.mosa.atr
winlog.event_data.IpAddress	83.97.115.19
winlog.event_data.IpPort	0
winlog.event_data.LogonGuid	{00000000-0000-0000-0000-00000000000}
winlog.event_data.ProcessId	0x614

Evidence:

Event ID: 4624 (Successful Logon).
Source IP: 83.97.115.19 (Potential External Threat Actor).
Account Name: wksadmin (Elevated Privileges).
Logon Type: Interactive Logon (Indicating Direct System Access).
Impact: The attacker successfully authenticated, likely using brute-forced credentials, leading to potential lateral movement and privilege escalation.

IMPACT (FILE CREATION & SYSTEM MODIFICATION)

SCREENSHOT: WINDOWS EVENT LOGS SHOWING SUSPICIOUS FILE CREATION AND SYSTEM MODIFICATIONS.

EVIDENCE:

Process: C:\Program Files (x86)\Java\jre8\bin\java.exe (Java execution may indicate malware deployment).

File Created: malicious.jar (Possible payload or persistence mechanism).

Timestamp: 11:30:38 AM (Aligns with attack progression). Impact: The attacker may have deployed malware, manipulated system files, or prepared data for exfiltration

Network logs showing outbound communication from 10.0.2.24 to an external IP. ***Evidence:***

Source IP: 10.0.2.24 (Internal System).

Destination IP: 204.79.197.200 (Potential External Server).

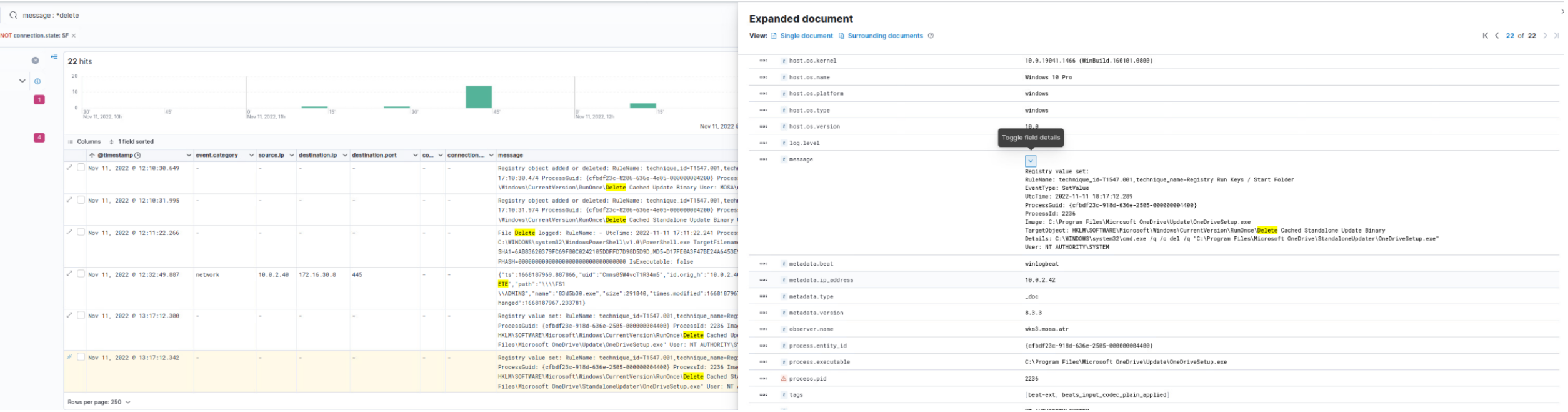
Geolocation: United States – Microsoft Corporation.

Network ASN: Microsoft (North America). Impact: Possible C2 beaconing or data exfiltration using a trusted service.

Network logs showing outbound communication from 10.0.2.24 to an external IP.

Impact: The attacker may be using Google DNS as a C2 mechanism or pivoting through another internal host.

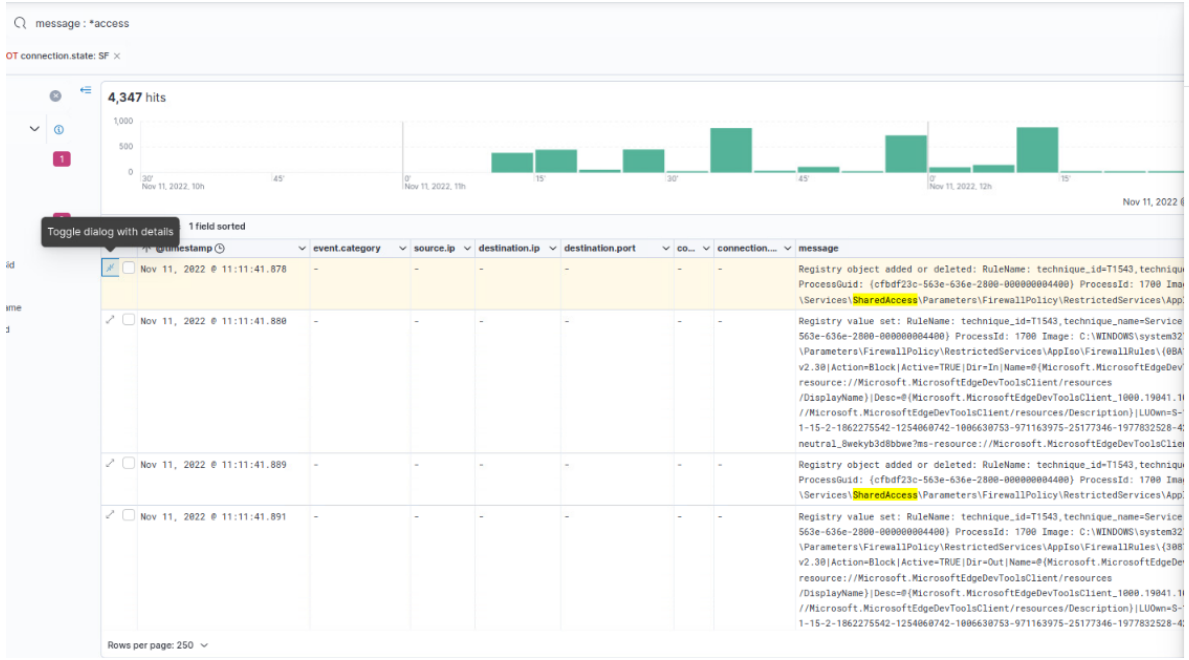
Next : Windows Event Logs showing a successful authentication event for a different account.

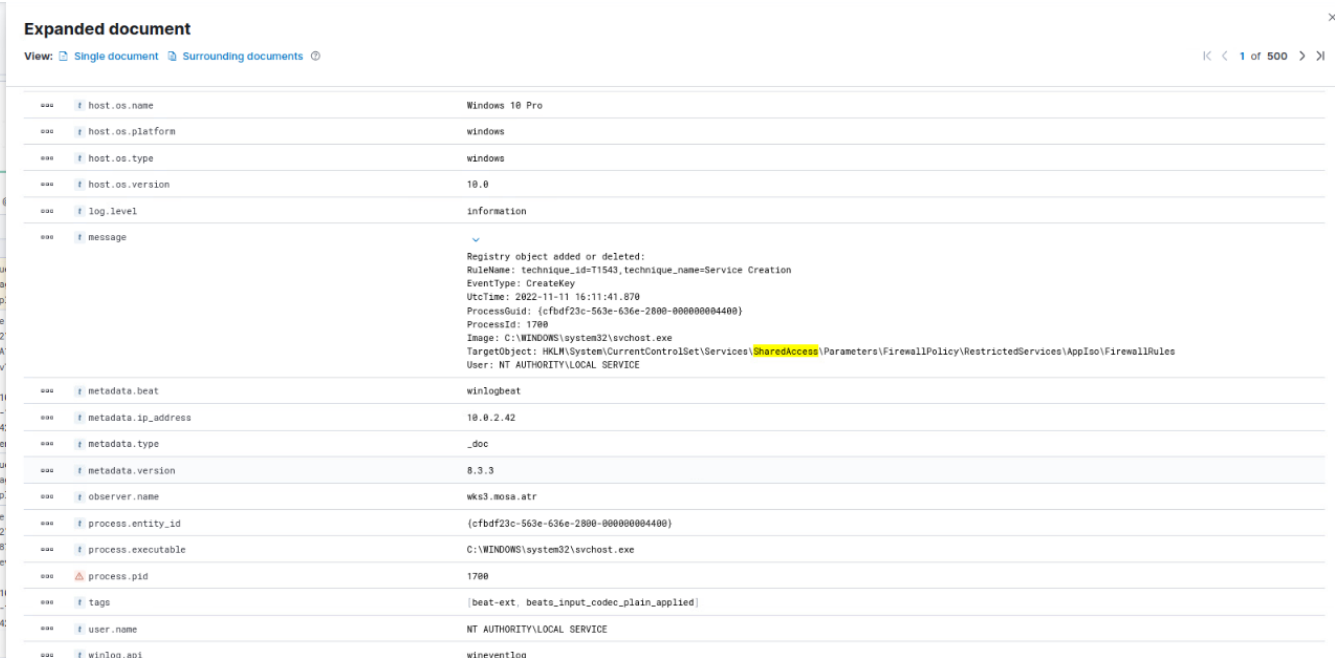


NOVEMBER 11, 2022 – MULTIPLE DELETION OF FILES (REGISTRY AND EXECUTABLE FILES)

IMPACT:

THIS ACTIVITY SUGGESTS POTENTIAL MALICIOUS BEHAVIOR RELATED TO PERSISTENCE, SCRIPT EXECUTION, AND SYSTEM MODIFICATION,



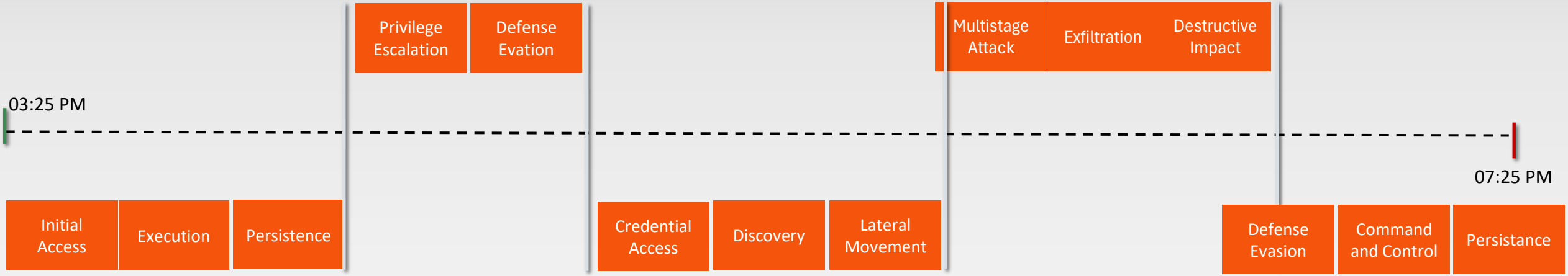


NOVEMBER 11, 2022 – FIREWALL POLICY MODIFICATION FOR STEALTH ATTACK. EVIDENCE:

REGISTRY KEY MODIFIED: HKLM\SYSTEM\CURRENTCONTROLSET\SERVICES\ (OFTEN USED FOR PERSISTENCE).

REGISTRY VALUE SET: STARTMODE (INDICATES SERVICE STARTUP BEHAVIOR, POTENTIALLY SET TO AUTO-START MALICIOUS SERVICES).

Incident Timeline based on MITRE Techniques (T1218)



Log Correlation

- Review and analyze the connections between the logs.

Indicator of Compromise

- Evading Detection
- Utilizing Task Scheduler, PowerShell, and Registry Modifications
- Payloads originating from .exe files and similar processes

Hypothesis

- System Binary Proxy Execution: Rundll32
- Similarities to the Leaving off the Land Binaries (LOLBin) Technique

Initial Access | Execution | Persistence

03:25 PM

A suspicious process, CompatTelRunner.exe, executed a PowerShell instance and performed unexpected reconnaissance on virtual machines.

03:30 PM

Credentials from the Credential Manager were accessed, indicating potential password theft. Sensitive cryptographic keys were accessed as well. Admin privileges have been assigned to the SYSTEM account. A key file operation showed signs of potential tampering.

03:31 PM

- ✓ Local group memberships have been enumerated.
- ✓ Admin privileges have been assigned to existing user (mpowel).
- ✓ Admin privileges are also assigned to the SYSTEM account.

03:32 PM

A guest account has logged in, which could indicate a potential backdoor. An admin login has also been detected.

03:34 PM

Suspicious registry modification detected. A cryptographic key has been exported, indicating a potential risk of credential theft.

03:38 PM

A registry modification has been detected, indicating a potential persistence mechanism.

03:47 PM

Regiexecuted PowerShell as a SYSTEM user with full control through a signed Microsoft binary (often referred to as LOLbins), which bypassed security tools.

There were multiple suspicious connections to the IP address 83.97.115.19 using rundll32.exe.

Additionally, there is a possible process injection involving the VMware tool to cmd.exe within web.mosa.atr. This activity affected the target user WEB\Administrator, and modifications were detected for users kneys and jsouthstry, indicating a potential persistence mechanism.

04:00 PM

Suspicious image detected: spssvc.exe, indicating a possible privilege escalation attempt. Firefox SQLite history has been deleted, suggesting potential evidence tampering.

Privilege Escalation

Defense Evation

04:00 PM

The initial malware was dropped on 172.64.155.188 from 10.0.2.41.

04:07 PM

1. Suspicious DNS query detected.
2. External HTTP connection identified.
3. Credential dumping activity detected.
4. File activity has been logged.

04:28 PM

1. Registry Modifications for Persistence
2. Cryptographic Operations Logged
3. User Account Modifications

04:35 PM

Privilege Escalation through Key Migration

04:45 PM

Attempt to Exfiltrate Data

04:50 PM

Multiple SSL connections to external IP addresses have been detected, along with a remote thread creation event.

04:55 PM

Lateral Movement Using Remote Desktop Protocol (RDP)

Credential Access

Discovery

Lateral Movement

04:55 PM

A connection attempt from an external source has been detected.

04:58 PM

Possible command and control (C2) communication detected.

05:06 PM

The execution of rundll32.exe.

05:15 PM

Suspicious DNS query to an unusual domain.

05:25 PM

There was an outbound connection made to a potentially suspicious external server (IP address: 3.222.199.124).

05:28 PM

Attempt to escalate privileges using the command `sc.exe`.

05:30 PM

Registry modification detected in the Run keys.

05:45 PM

Multiple SSL connections to external IPs

05:55 PM

A potential RDP session has been initiated.

06:00 PM

Multistage Attack | Exfiltration | Destructive Impact

6:00 PM

Initial activity has been observed.

6:20 – 6:30 PM

There are multiple files being created that are associated with svchost.exe, which may suggest process injection or suspicious modifications to files.

6: 40 PM

High-frequency activity was detected, indicating repeated changes to file system permissions, which suggests attempts at privilege escalation.

6:45 PM

Concerns arise when svchost.exe interacts with system files, indicating possible process masquerading or unauthorized execution.

6:50 – 6:54 PM

There has been an increase in system interaction, which may indicate potential data exfiltration, including access to sensitive directories.

6: 55 PM

Defense Evasion

Command and Control

Persistance

6:57 PM – 7:15 PM

There are multiple DNS inquiries from the internal host, which could indicate potential manipulation of DNS queries.

Noticed logs indicating files were deleted, suggesting potential malicious activity.

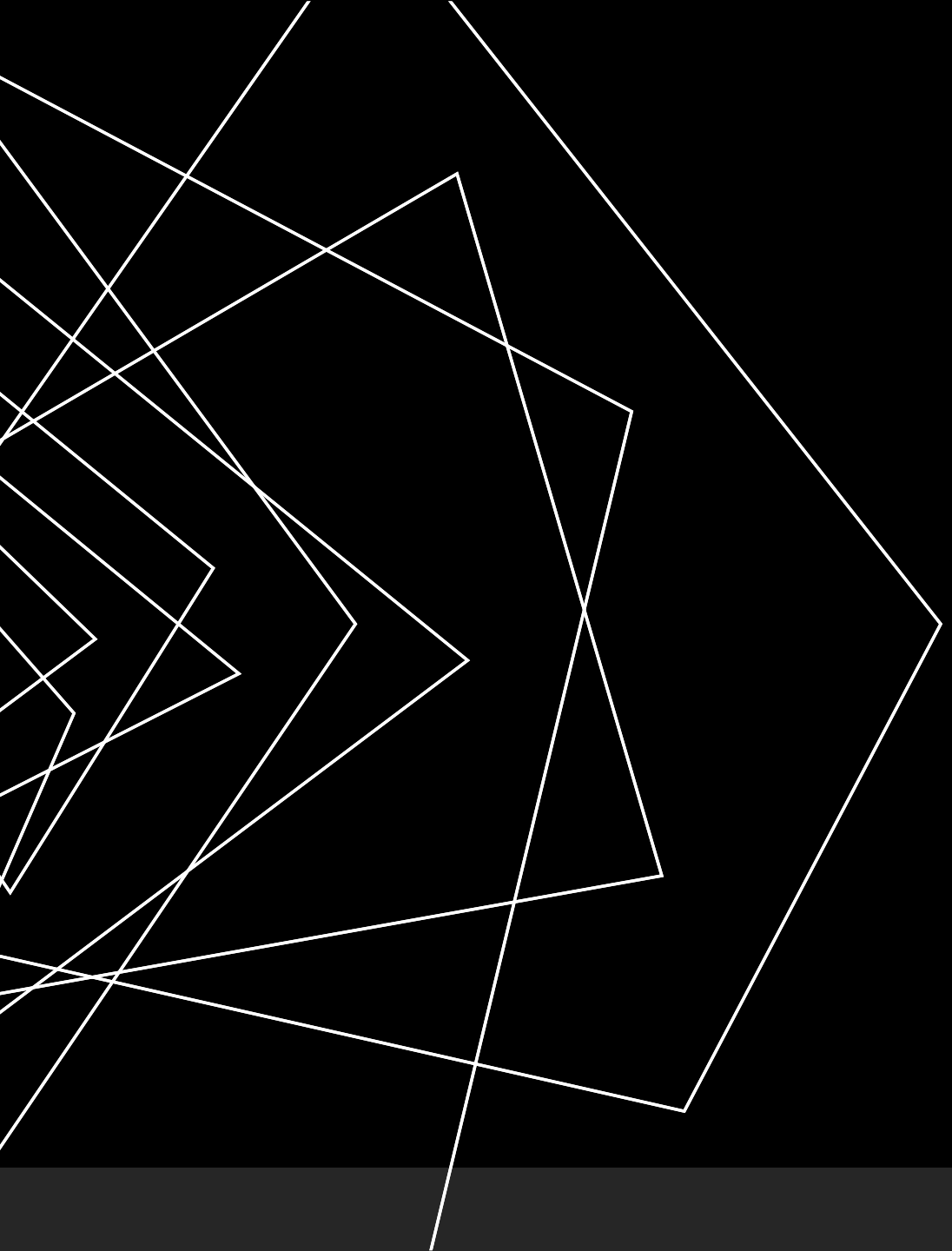
7:02 PM - 7:10 PM

7:15 PM – 7:25 PM

Multiple instances of svchost.exe have been created, which could potentially be malicious files or payloads.

Options for Registry Key Values

7:23 PM - 7:25 PM



Thank you